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(54) **SYSTEMS AND METHODS FOR
AUTOMATED TELLER DRAWER
COUNTING AND BALANCING**

(71) Applicant: **Wells Fargo Bank, N.A.**, San Francisco, CA (US)

(72) Inventors: **Alicia Y. Moore**, San Ramon, CA (US);
Miranda C. Hill, Seattle, WA (US);
Susan Symons, Mooresville, NC (US)

(73) Assignee: **Wells Fargo Bank, N.A.**, San Francisco, CA (US)

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See application file for complete search history.

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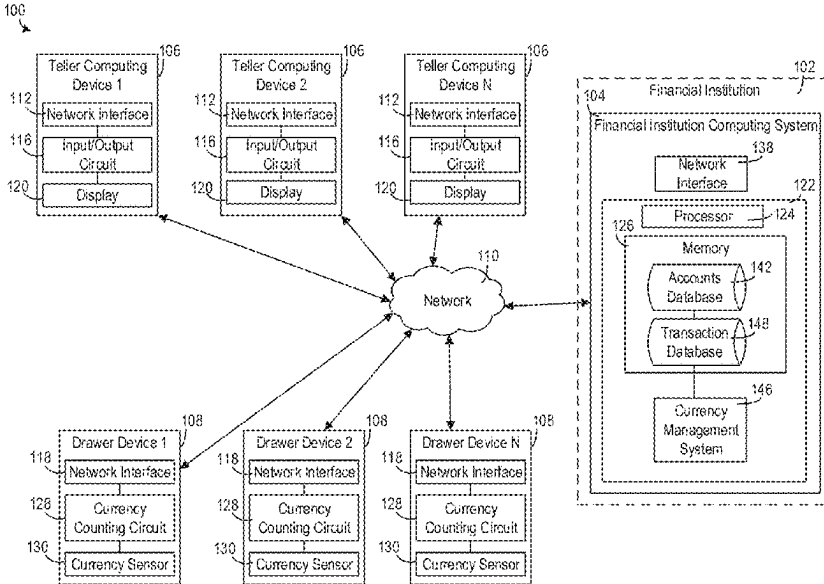
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Primary Examiner — Rokib Masud
(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

(57) **ABSTRACT**

A system includes a currency counting device communicably coupled to memory and one or more processors. The memory stores instructions that are executable by the one or more processors to cause the one or more processors to cause a first amount of currency to be counted by the currency counting device, wherein the first amount of currency is an actual amount of currency received by a teller money drawer; determine that the first amount of currency is a discrepancy amount different than an expected amount of currency; determine that a teller error is a potential source of the discrepancy based on the discrepancy amount being below a threshold; and cause a display device of a teller computing device to display an indication within a pop-up window that the teller error is the potential source of the discrepancy based on the discrepancy amount being below the threshold.

20 Claims, 4 Drawing Sheets



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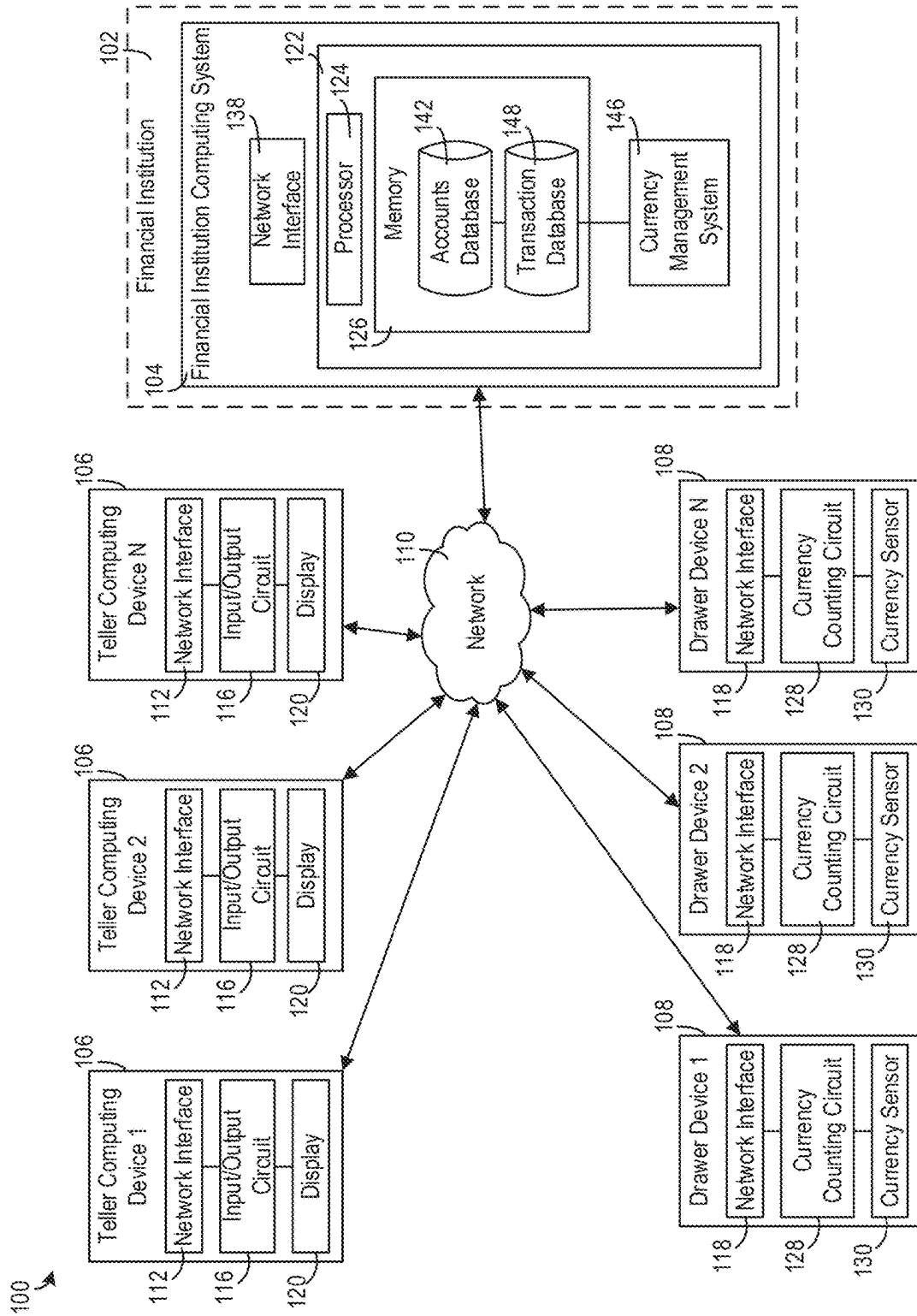


FIG. 1

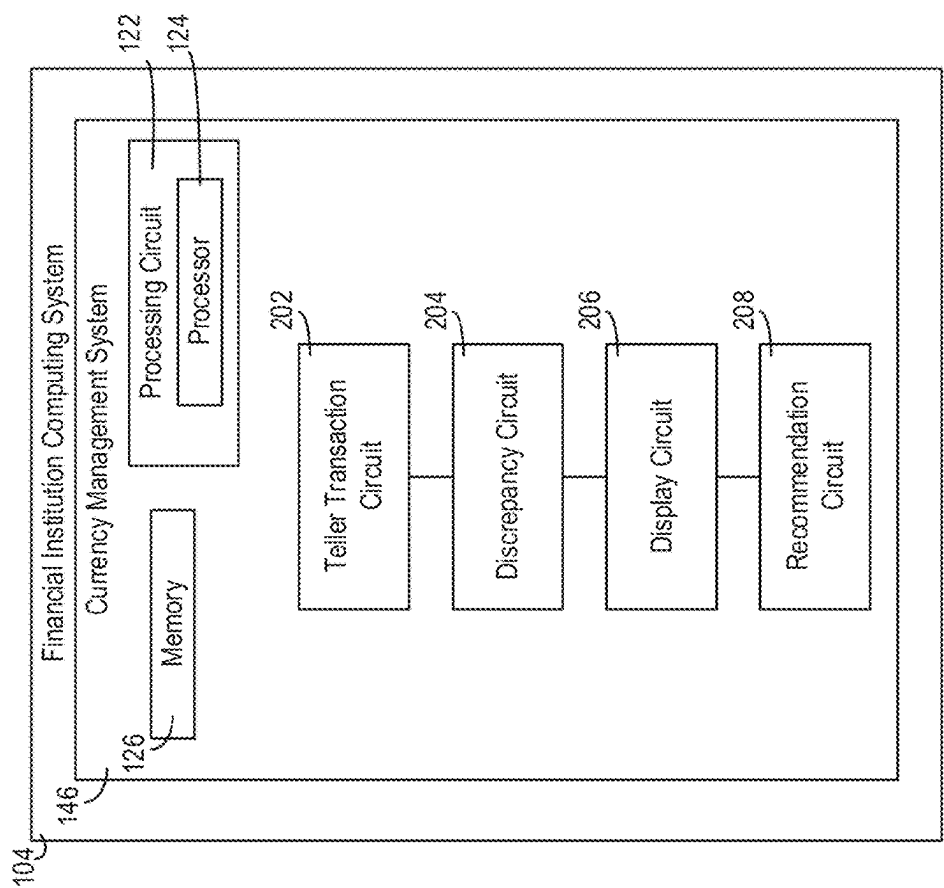


FIG. 2

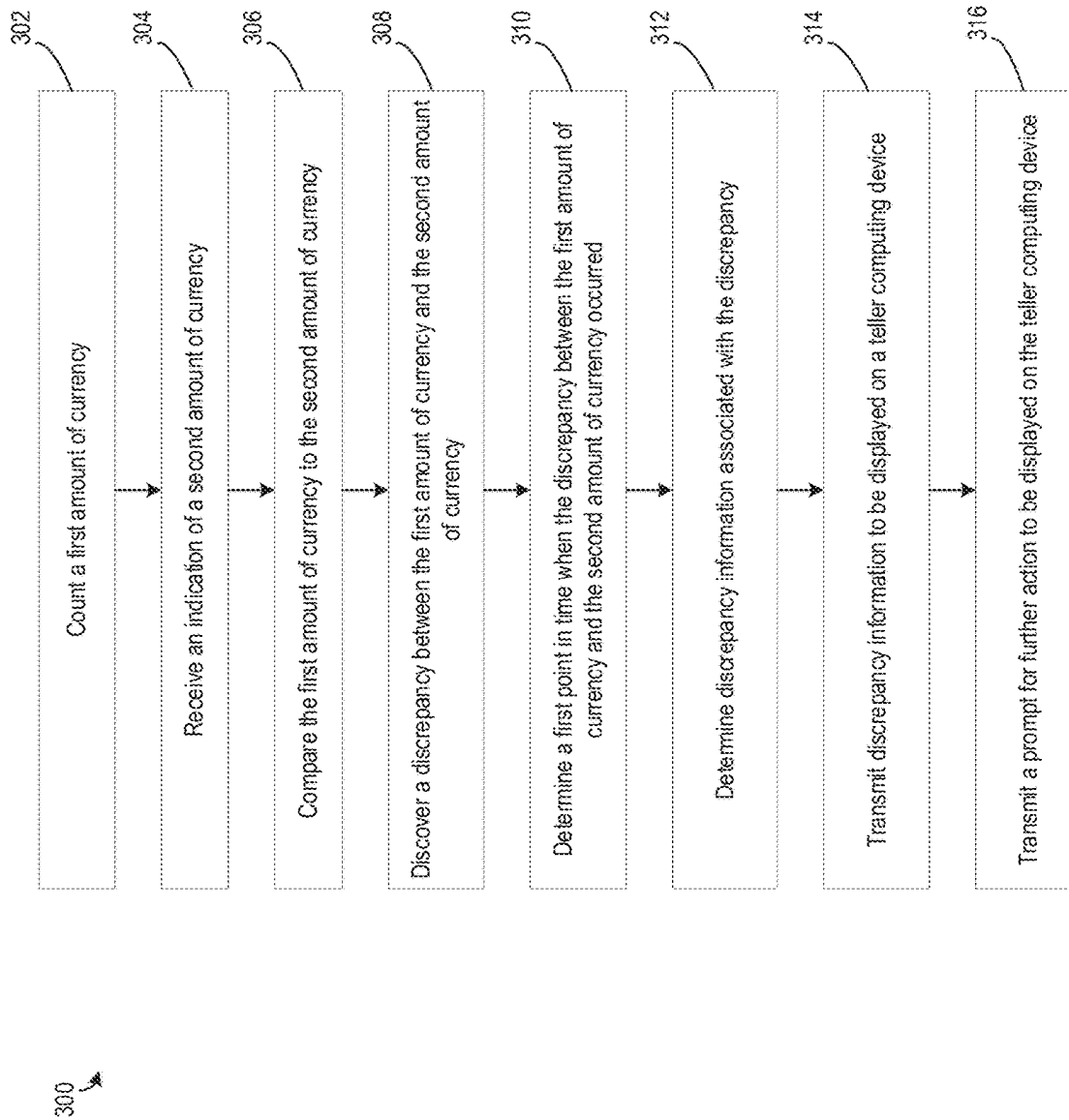


FIG. 3

400

Currency Management System					
Teller Drawer Balance Information					
402	404	406	408	410	
Drawer	Expected Currency Amount	Actual Currency Amount	Suspected Transaction No.	Suspected Time	
Drawer 1	\$1115	\$1005	001345	1:05pm CST	
Drawer 2	\$1100	\$1210	012654	3:24pm CST	

FIG. 4

500

Currency Management System	
Teller Drawer Balance Information	
Recommended Action	
502	Drawer 1
504	1. Check Drawer 2 for Surplus 2. Manual Recount

FIG. 5

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SYSTEMS AND METHODS FOR AUTOMATED TELLER DRAWER COUNTING AND BALANCING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/367,042, filed Jul. 2, 2021, which is a continuation of U.S. patent application Ser. No. 15/388,008, filed Dec. 22, 2016, which claims priority to U.S. Provisional Patent Application No. 62/317,389, filed Apr. 1, 2016, which are each incorporated herein by reference in their entireties.

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to the field of currency counting and balancing.

BACKGROUND

Financial institutions desire to efficiently and accurately determine when discrepancies in the expected currency and the actual currency in teller drawers and/or vaults occur. Traditionally, at the end of a shift, tellers at financial institutions reconcile the teller drawers with the transactions that were processed during their respective shift. In a conventional system, the tellers must recount the currency in the teller drawers by hand, having to recount each time a discrepancy between the actual amount of currency and the expected amount of currency is found. Counting and recounting currency by hand is inaccurate and inefficient and thus, in addition to wasting time, currency discrepancies may never be discovered.

SUMMARY

A first example embodiment relates to a system. The system includes a currency counting device and a processing circuit. The processing circuit is communicably coupled to the currency counting device. The processing circuit includes a processor and memory. The memory is structured to store instructions that are executable by the processor and cause the processing circuit to cause a first amount of currency to be counted by the currency counting device, wherein the first amount of currency is the actual amount of currency received by a teller money drawer, receive an indication of a second amount of currency, wherein the second amount of currency is an expected amount of currency, compare the first amount of currency to the second amount of currency, discover a discrepancy between the first amount of currency and the second amount of currency, and transmit discrepancy information to be displayed on a teller computing device.

Another example embodiment relates to a method. The method includes counting, by a currency counting circuit, a first amount of currency, wherein the first amount of currency is the actual amount of currency received by a teller money drawer, receiving, by a teller transaction circuit, an indication of a second amount of currency, wherein the second amount of currency is an expected amount of currency, comparing, by a discrepancy circuit, the first amount of currency to the second amount of currency, discovering, by the discrepancy circuit, a discrepancy between the first amount of currency and the second amount of currency, and

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transmitting, by a display circuit, discrepancy information to be displayed on a teller computing device.

A further example embodiment relates to an apparatus. The apparatus includes a currency counting circuit structured to count a first amount of currency, wherein the first amount of currency is the actual amount of currency received by a teller money drawer, a teller transaction circuit structured to receive an indication of a second amount of currency, wherein the second amount of currency is an expected amount of currency, a discrepancy circuit structured to compare the first amount of currency to the second amount of currency, and discover a discrepancy between the first amount of currency and the second amount of currency, a display circuit structured to transmit discrepancy information to be displayed on a teller computing device.

These and other features, together with the organization and manner of operation thereof, will become apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a schematic diagram of a currency management system, according to an example embodiment.

FIG. 2 is a diagram of the currency management system of FIG. 1, according to an example embodiment.

FIG. 3 is a flow diagram of a method of determining a currency discrepancy using the currency management system of FIGS. 1-2, according to an example embodiment.

FIG. 4 is a diagram of a display generated by the currency management system of FIGS. 1-2, according to an example embodiment.

FIG. 5 is a diagram of another display generated by the currency management system of FIGS. 1-2, according to an example embodiment.

DETAILED DESCRIPTION

Referring to the Figures generally, various systems, methods, and apparatuses for facilitating currency management at a financial institution are described herein. More particularly, systems and methods for facilitating counting and reconciliation of discrepancies in the expected amount of currency and the actual amount of currency in a teller drawer are described herein.

According to various example embodiments, as described in further detail below, facilitating management of currency in a financial institution in an automated system may improve the ability to count the currency more accurately and efficiently than is done with a manual counting and balancing system. Unlike conventional currency management, among various other functionality, the embodiments described herein generate an analysis of when and during which transaction a currency discrepancy may have taken place. Further, the currency management system described herein may notify a teller of a discrepancy using a display of a teller computing device and may additionally recommend further action based on the discrepancy information identified.

An example implementation may be described as follows. A customer enters a financial institution (e.g., a bank branch) to conduct a transaction. During the transaction, a teller working at the financial institution receives currency from and/or gives currency to the customer based on the type of transaction (e.g., deposit, withdrawal, transfer). The teller deposits the currency into or takes the currency from a teller drawer, herein referred to as a drawer device. Concurrently,

the teller enters transaction information into a teller computing device, which then stores the transaction information into the transaction database located with the financial institution computing system. After a set period of time (e.g., one shift, one hour, one day) and often after a plurality of transactions have occurred, the teller reconciles the drawer device with the transactions that are stored and available for review on the teller computing device. Specifically, the teller reconciles the drawer device to ensure that the actual amount of funds in the drawer device matches an expected amount of funds in the drawer device. Rather than count the currency in the drawer device manually, the contemplated currency management system monitors the drawer device at all times to identify any potential discrepancies between what is noted in the transaction database and the amount of currency contained within the drawer device. Beneficially, any discrepancies may be known contemporaneously or substantially contemporaneously to any mismatch occurring. Accordingly, upon detection of a problem, the currency management system generates and provides a message to the teller computing device to display the potential discrepancy and any related information about the discrepancy.

In operation, the currency management system facilitates the real-time display of currency discrepancies for each drawer device at a financial institution. Each teller is responsible for a particular drawer device such that each teller may receive displays relating to that device. Other personnel, such as managers at the financial institution, may receive notifications regarding all of the drawer devices to monitor the activity of the tellers and each of the drawers. Other embodiments may include different arrangements and displays without departing from the spirit and scope of the present disclosure.

Referring now to FIG. 1, a block diagram of a system 100 is shown, according to an example embodiment. As described in further detail below, the system 100 facilitates currency management in a financial institution 102. Specifically, the system 100 facilitates the determination of discrepancy information relating to transactions in a financial institution 102. As used herein, the term “currency” refers to fiat currency, such as paper money and coin money that is declared by a government to be legal tender (e.g., US Dollars, Canadian Dollars, Chinese Yuan, Euros, Japanese Yen, etc.). In some arrangements, the system 100 additionally facilitates the recommendation of further action when a discrepancy is determined to have occurred. The system 100 receives an amount of currency during a transaction between a customer and a teller at the financial institution 102. The system 100 then counts the currency, compares the actual amount of currency to the expected amount of currency, and facilitates the determination of discrepancy information. Further, the system 100 recommends actions to be taken based on the determination of discrepancy information, such as a manual recount of the teller drawer.

As shown, the system 100 includes a financial institution computing system 104 associated with a financial institution 102, the financial institution computing system 104 is communicably and operatively coupled to one or more teller computing devices 106 and one or more drawer devices 108 over a network 110. The network 110 provides communicable and operative coupling between the teller computing devices 106, drawer devices 108, and the financial institution computing system 104, and other components disclosed and described herein to provide and facilitate the exchange of communications (e.g., data, instructions, messages, values, commands, etc.) Accordingly, the network 110 may include any network including wired (e.g., Ethernet) and/or

wireless networks (e.g., 802.11X, ZigBee, Bluetooth, WiFi, etc.). In some arrangements, the network 110 includes the Internet. In further embodiments, the network 110 includes a proprietary banking network to provide secure or substantially secure communications.

As shown, the system 100 depicts multiple teller computing devices 106 (e.g., teller computing device 1, teller computing device 2, etc.) and multiple drawer devices 108 (e.g., drawer device 1, drawer device 2, etc.). This depiction is for illustrative purposes only to show an implementation environment of the systems and methods described herein. Each of these entities may have the same or similar characteristics. For the purpose of clarity, the disclosure contained herein is in reference to a single teller computing device and a single drawer device.

The teller computing device 106 includes any type of computing device that may be used to facilitate management of currency within the financial institution 102. In some arrangements, the teller uses the teller computing device 106 to assist customers of the financial institution 102 with in-person transactions. In this regard, the teller computing device 106 may include any wearable and non-wearable device. Wearable devices refer to any type of device that an individual wears including, but not limited to, a watch (e.g., smart watch), glasses (e.g., eye glasses, sunglasses, smart glasses, etc.), bracelet (e.g., a smart bracelet), etc. Teller computing device 106 may also include any type of mobile device including, but not limited to, a phone (e.g., smart phone, etc.) and/or computing devices (e.g., desktop computer, laptop computer, personal digital assistant, etc.).

The teller computing device 106 further includes a display 120, an input/output circuit 116, and network interface 112. The network interface 112 of the teller computing device 106 is adapted for and configured to establish a communication session via the network 110 with the financial institution computing system 104 and the drawer device 108. Accordingly, the network interface 112 includes any of a cellular transceiver (Code Division Multiple Access (CDMA), Global System for Mobile Communications (GSM), Long-Term Evolution (LTE), etc.), a wireless network transceiver (e.g., 802.11X, ZigBee, Bluetooth, etc.), or a combination thereof (e.g., both a cellular transceiver and a Bluetooth transceiver). In some embodiments, the network interface 112 communicates via a secured wired connection with a branch of the financial institution 102.

The display 120 is used to present account information, transaction information, and the like to tellers on the teller computing device 106. In this regard, the display 120 is communicably and operatively coupled to the input/output circuit 116 to provide a user interface for receiving and displaying information on the teller computing device 106. Examples of user interfaces are described more fully herein with regard to FIGS. 4 and 5.

The input/output circuit 116 is structured to receive and provide communication(s) to a user (e.g., a bank teller) of the teller computing device 106 (or, to another entity such as the drawer device 108). In this regard, the input/output circuit 116 is structured to exchange data, communications, instructions, etc. with an input/output component of the teller computing device 106. Accordingly, in one embodiment, the input/output circuit 116 includes an input/output device such as a display device, a touchscreen, a keyboard, and a microphone. In another embodiment, the input/output circuit 116 may include communication circuitry for facilitating the exchange of data, values, messages, and the like between an input/output device and the components of the teller computing device 106. In yet another embodiment, the

input/output circuit **116** may include machine-readable media for facilitating the exchange of information between the input/output device and the components of the teller computing device **106**. In still another embodiment, the input/output circuit **116** may include any combination of hardware components (e.g., a touchscreen), communication circuitry, and machine-readable media.

The drawer device **108** may include any type of storage device that may be used by a teller, or other financial institution personnel, to store and access currency at a financial institution **102**. In this regard, the drawer device **108** may include any sliding-type teller drawer used to store currency, which can be locked and unlocked by the teller during a transaction, and may also include vault-type storage, which may be used to store higher denominations of currency which require a more secure type of storage.

The drawer device **108** further includes a currency counting circuit **128**, a currency sensor **130**, and network interface **118**. The network interface **118** of the drawer device **108** is adapted for and configured to establish a communication session via the network **110** with the financial institution computing system **104** and the teller computing device **106**. Accordingly, the network interface **118** includes any of a cellular transceiver (Code Division Multiple Access (CDMA), Global System for Mobile Communications (GSM), Long-Term Evolution (LTE), etc.), a wireless network transceiver (e.g., 802.11X, ZigBee, Bluetooth, etc.), or a combination thereof (e.g., both a cellular transceiver and a Bluetooth transceiver). In some embodiments, the network interface **118** communicates via a secured wired connection within a branch of the financial institution **102**.

The currency sensor **130** is structured to receive the currency deposited into the drawer device **108**, count the currency, and communicate the amount of currency to the currency counting circuit **128**. In this regard, the currency sensor **130** is communicably and operatively coupled to the currency counting circuit **128** to facilitate the process. The currency sensor **130** includes any type of sensor to count the currency. In some embodiments, the currency sensor **130** includes paper money counters structured to count paper money by mechanically pulling each bill through the sensor and by counting the number bills and types of bills (e.g., by scanning identifying codes, by interpreting the number of times a beam of light is interrupted, by comparing an image of each bill to pattern recognition criteria, etc.). Accordingly, in some embodiments, the currency sensor **130** distinguishes between the denominations of bills and/or the type of currency (e.g., US Dollars, Canadian Dollars, etc.). In some embodiments, the currency sensor **130** includes coin counters structured to both sort and count coins simultaneously (e.g., by scanning identifying codes, by interpreting the number of times a beam of light is interrupted, by comparing an image of each coin to pattern recognition criteria, etc.). In further embodiments, the currency sensor **130** is additionally equipped to distinguish genuine currency from counterfeit currency.

The currency counting circuit **128** is structured to receive the counted currency amount from the currency sensor **130** and communicate that information to the teller computing device **106**. In this regard, the currency counting circuit **128** is communicably and operatively coupled to the currency sensor **130** and the teller computing device **106**. In one embodiment, the currency counting circuit **128** includes various hardware components, such as a transmitter, to facilitate the sending of currency amount information to the currency management system **146** via the network **110**. In another embodiment, the currency counting circuit **128**

includes communication circuitry including, but not limited to, wired and wireless communication protocol to facilitate transmission of the currency amount information from the currency sensor **130** to the currency management system **146**. In yet another embodiment, the currency counting circuit **128** includes any combination of hardware components (e.g., a transmitter) and communication circuitry.

Still referring to FIG. **1**, the system **100** includes the financial institution computing system **104**. The financial institution computing system **104** includes a network interface **138**, which is used to establish connections with other components of the system **100** by way of network **110**. The network interface **138** includes program logic that facilitates connection of the financial institution computing system **104** to the network **110**. The network interface **138** supports communication between the financial institution computing system **104** and other systems, such as the teller computing device **106** and the drawer device **108**. For example, the network interface **138** includes a cellular modem, a Bluetooth transceiver, a Bluetooth beacon, a radio-frequency identification (RFID) transceiver, and a near-field communication (NFC) transmitter. In some embodiments, the network interface **138** communicates via a secured wired connection within a branch of the financial institution **102**. In some arrangements, the network interface **138** includes the hardware and machine-readable media sufficient to support communication over multiple channels of data communication. Further, in some arrangements, the network interface **138** includes cryptography capabilities to establish a secure or relatively secure communication session with the financial institution computing system **104**, teller computing device **106**, and drawer device **108**. In this regard, financial data (or other types of data) may be encrypted and transmitted to prevent or substantially prevent the threat of hacking.

The financial institution computing system **104** further includes an accounts database **142**. The accounts database **142** is configured to hold, store, categorize, and otherwise serve as a repository for information associated with accounts held by the financial institution **102**. For example, the accounts database **142** may store account numbers, account balances, account ownership information, and the like. The accounts database **142** is structured to selectively provide access to information relating to an account at the financial institution **102**. In this regard, as discussed further herein, the accounts database **142** is communicably and operatively coupled to the currency management system **146** to provide access to such information, such that the currency management system **146** may facilitate the determination of discrepancies in actual currency and expected currency at the financial institution **102** and recommend further action based on that information.

The financial institution computing system **104** further includes a transaction database **148**. The transaction database **148** is configured to hold, store, categorize, and otherwise serve as a repository for transaction information. The transaction database **148** is structured to selectively provide access to information relating to one or more transactions that occur at a financial institution **102**. In this regard, the transaction database **148** is communicably and operatively coupled to the currency management system **146** to provide access to such information, such that the currency management system **146** may use that information to determine during which transaction a currency discrepancy has taken place. Such processes are described more fully herein below.

As used herein, "transaction information" may include any information that a teller at a financial institution **102**

may enter on a teller computing device **106** during a transaction process with a customer. In some arrangements, the transaction information includes an amount of currency deposited into or an amount of currency withdrawn from an account at the financial institution, the time of the transaction, the account number involved with the transaction, the name of the account holder, and a unique transaction number that can be used to identify the particular transaction.

The financial institution computing system **104** includes a currency management system **146** for managing a currency counting and balancing process at the financial institution **102**. The currency management system **146** is structured to count the accepted currency, receive transaction information including the amount of currency deposited or withdrawn from an account, and discover discrepancies between the actual amount of currency in a teller drawer and the expected amount of currency. In some embodiments, the currency management system **146** uses the transaction information to recommend further action to be taken by a teller when a discrepancy is discovered. The currency management system **146** may be configured to use transaction and account information stored in the transaction database **148** and accounts database **142**, respectively, to determine which options and/or recommendations to provide to a teller when a discrepancy is discovered, according to an example embodiment. Further, in some arrangements, the currency management system **146** is configured to transmit a message for display to a teller computing system reflecting recommendations for further action.

Referring now to FIG. 2, a diagram of the currency management system **146** and part of the financial institution computing system **104** is shown according to an example embodiment. As shown, financial institution computing system **104** includes a processing circuit **122** having a processor **124** and a memory **126**. The processor **124** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The one or more memory devices **126** (e.g., RAM, NVRAM, ROM, Flash Memory, hard disk storage, etc.) may store data and/or computer code for facilitating the various processes described herein. Moreover, the one or more memory devices **126** may be or include tangible, non-transient volatile memory or non-volatile memory. Accordingly, the one or more memory devices **126** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein.

The currency management system **146** may be embodied with the financial institution computing system **104**. Accordingly, in some arrangements, the currency management system **146** may be embodied or at least partly embodied in the memory **126**, where at least some operations may be executable from the processing circuit **122**. As described above, the currency management system **146** facilitates counting and balancing currency in a drawer device **108** at the financial institution **102**. The currency management system **146** determines when and during which transactions discrepancies in the expected and actual currency occur for a drawer device **108**. Beneficially, such a system may facilitate efficient and accurate determinations of how and when discrepancies in currency at the financial institution **102** occur.

The currency management system **146** is shown to include a teller transaction circuit **202**, a discrepancy circuit

204, a display circuit **206**, and a recommendation circuit **208**, with all such circuits communicably coupled to each other. Other embodiments may include more or less circuits without departing from the spirit and scope of the present disclosure. Further, some embodiments may combine the activities of one circuit with another circuit to form a single circuit. Therefore, those of ordinary skill in the art will appreciate that the present arrangement is not meant to be limiting.

The teller transaction circuit **202** is structured to receive information from the transaction database **148** regarding transaction information from transactions that occur at the financial institution **102**. The teller transaction circuit **202** is additionally structured to receive information from a drawer device **108** regarding the amount of currency that has been counted by the currency sensor **130**. Further, the teller transaction circuit **202** is structured to communicate the received information to the discrepancy circuit **204**. In this regard, in one embodiment, the teller transaction circuit **202** is communicably and operatively coupled to the transaction database **148** to receive the information regarding an expected amount of currency and to a drawer device **108** to receive information regarding an actual amount of currency that is in the drawer device **108**. In further embodiments, the teller transaction circuit **202** is additionally communicably and operatively coupled to the discrepancy circuit **204** to communicate the transaction information and currency counting information to the discrepancy circuit **204** for evaluation and discrepancy identification.

The discrepancy circuit **204** is structured to identify potential discrepancies in currency and determine when they may have occurred. The discrepancy circuit **204** compares an expected amount of currency to an actual amount of currency contained in a drawer device **108**. The discrepancy circuit **204** receives information regarding the actual and expected amounts of currency from the teller transaction circuit **202** as mentioned above. This information may relate to a set time period, such as receiving transaction information for transactions occurring during a specific teller shift, hour, or day, etc. In some embodiments, the discrepancy circuit **204** is further structured to communicate the identified discrepancies and information relating to the discrepancies (e.g., transaction information, account information) to the display circuit **206** so that the display circuit **206** generates a message to be displayed on a teller computing device **106**. In this regard, the discrepancy circuit **204** may be communicably and operatively coupled to the teller transaction circuit **202** and to the display circuit **206**.

The display circuit **206** is structured to generate and provide a message to the teller computing device **106** to display one or more sets of discrepancy information identified by the discrepancy circuit **204**. In some arrangements, the display circuit **206** is also structured to generate and provide a message to the teller computing device **106** regarding possible further action that the teller may need to take. In this regard, the display circuit **206** may be communicably and operatively coupled to the discrepancy circuit **204** and the recommendation circuit **208**.

The display circuit **206** is configured to create, generate, establish, update, and maintain a discrepancy list of all identified potential discrepancies and any information associated therewith. Accordingly, in one embodiment, the display circuit **206** includes a list generating tool. In another embodiment, the display circuit **206** includes communication circuitry for facilitating the exchange of information between and among the display circuit **206** and any other circuitry or logic. In yet another embodiment, the display

circuit **206** includes any combination of machine-readable media, list generating tool, and communication circuitry. In further embodiments, the display includes a list of discrepancy information, wherein a teller (or any user of the teller computing device **106** may observe or search the list for information associated with the identified discrepancies. Information included in the list may include, but is not limited to, the drawer device **108** from which the amount of actual currency was counted, an identification of a transaction number associated with the discrepancy, the identified time that the discrepancy occurred, expected currency amount, and the actual currency amount.

The recommendation circuit **208** is structured to provide at least one recommended action based on the discrepancy information identified for a particular drawer device **108**. In some arrangements, the recommended action is provided via the display **120** on the teller computing device **106**. In this regard, the recommendation circuit **208** is communicably and operatively coupled to the display circuit **206** to facilitate the display of the recommended action on a teller computing device **106**. The information provided with a recommended action may specify the drawer device **108** where the currency discrepancy was identified and at least one recommended action. As an example of a recommended action, the recommendation circuit **208** provides a recommendation for a “manual recount” of the drawer **1** contents. As another example, if there is a discrepancy in both drawer **1** and drawer **2** and the amount in drawer **1** is \$100 less than expected and drawer **2** is \$100 more than expected, then the recommended action states to “check drawer **2** for surplus.”

Referring now to FIG. **3**, an example method of managing, counting, and balancing currency in a drawer device at a financial institution is depicted. Method **300** may be implemented by the currency management system **146** of FIGS. **1-2**, such that reference may be made to one or more components of FIGS. **1-2** in explaining method **300**.

A first amount of currency is counted at **302**. In some arrangements, the currency management system **146** counts the first amount of currency contained within a drawer device **108** (e.g., through the currency counting circuit **128** and the currency sensor **130**). As mentioned above, counting currency within the drawer device **108** may be performed using a currency sensor **130**. As noted, the currency sensor **130** may count paper money by either electrical or mechanical means, or by employing both mechanical and electrical functionality. The currency sensor **130** may additionally be equipped to distinguish denominations and/or types of currency. In addition to counting paper money, the currency sensor **130** may include a coin counter. Furthermore, a currency counterfeit determination mechanism may be used in combination with the currency sensor **130**. Once the currency sensor **130** has counted the currency (including coins and paper money), the currency counting circuit **128** receives the first amount of currency. As noted, the first amount of currency is the actual amount of currency in the drawer device **108** that is compared to the expected amount of currency (referred to as the “second amount of currency”).

An indication of a second amount of currency is received at **304**. In some arrangements, the teller transaction circuit **202** receives the indication of the second amount of currency. As mentioned above, the teller transaction circuit **202** may receive transaction information from the transaction database **148**. The transaction information may include the second amount of currency (e.g., the expected amount of currency), along with other information relating to the transaction between the customer and the teller at the

financial institution. For example, when a transaction occurs at the financial institution, the customer may deposit a certain amount of money (e.g., \$100) and the teller may enter into the teller computing device **106** that the amount of money has been deposited by the customer, thus denoting the expected amount of currency. In some embodiments, the second amount of currency may include a total netted amount of expected currency at the end of a set time period (e.g., at the end of a teller shift, end of each hour, end of each day).

The first amount of currency is compared to the second amount of currency at **306**. In some arrangements, the discrepancy circuit **204** compares the first amount of currency to the second amount of currency. As mentioned above, the discrepancy circuit **204** may receive transaction information from the teller transaction circuit **202** indicating a second (or expected) amount of currency and a first (or actual) amount of currency. The discrepancy circuit **204** may then compare the amounts to identify a discrepancy between the amounts.

A discrepancy between the first amount of currency and the second amount of currency is discovered at **308**. The discrepancy circuit **204** discovers the discrepancy between the first amount of currency and second amount of currency. As mentioned, using the transaction information received from the teller transaction circuit **202**, the discrepancy circuit **204** may perform an analysis to determine that a discrepancy exists between the expected and actual amounts of currency. A discrepancy between the first amount of currency and the second amount of currency includes that the amounts do not match (e.g., a first amount of currency is \$100 and a second amount of currency is \$1000).

A first point in time when the discrepancy between the first amount of currency and the second amount of currency occurred is determined at **310**. In some arrangements, the discrepancy circuit **204** determines the first point in time when the discrepancy occurred. As mentioned above, the currency management system **146** may be communicably and operatively coupled to the transaction database **148**. The transaction database **148** stores information regarding the transactions that take place at the financial institution. Specifically, the transaction database **148** may store information regarding the transaction numbers, the time at which the transactions took place, and the amounts of currency deposited or withdrawn during those transactions. As noted above, the transaction information received from the transaction database **148** includes information relating to a set time period, such as receiving transaction information for transactions occurring during a specific teller shift, hour, or day, etc. The discrepancy circuit **204** may additionally receive information regarding transactions on a time-period basis (e.g., a sweep of data may occur every teller shift change, every hour, every few hours, etc.) from the teller transaction circuit **202**. By taking into consideration the transaction information including, but not limited to, the transaction times and amounts of currency deposited and/or withdrawn, the discrepancy circuit **204** identifies a point in time at which there was first a discrepancy between the expected amount of currency and the actual amount of currency in a particular drawer device **108**.

Discrepancy information associated with the discrepancy is determined at **312**. In some arrangements, the discrepancy circuit **204** determines further discrepancy information. Similar to process **310**, the discrepancy circuit **204** may be in communication with the teller transaction circuit **202** to receive transaction information. In some embodiments, part of this transaction information includes the account infor-

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mation relating to the transaction during which the discrepancy potentially occurred. In this regard, detailed account information is additionally received by the teller transaction circuit 202 from the accounts database 142. Furthermore, in some embodiments, the discrepancy circuit 204 uses trans-
 action information to determine errors in teller data entry. For example, the teller accepts a \$1000 deposit and incor-
 rectly types into the teller computing device 106 that \$100 was deposited, resulting in a typographical mistake. In yet
 another embodiment, the discrepancy circuit 204 determines
 other potential sources of teller errors. For example, the
 teller may drop a coin and/or bill or may deliver more
 currency to a customer than is due. In some embodiments,
 the discrepancy circuit 204 determines that such a teller
 error may be the source of the discrepancy if the discrepancy
 is not a large amount. In other embodiments, the discrepancy
 circuit 204 does not use the discrepancy between the
 expected and actual currency to determine that teller error is
 the potential source of the discrepancy. Thus, in some
 arrangements, the discrepancy circuit 204 uses all informa-
 tion received from the transaction database 142 to determine
 further discrepancy information.

Discrepancy information to be displayed on a teller com-
 puting device is transmitted at 314. In some arrangements,
 the display circuit 206 transmits the discrepancy information
 to be displayed on the teller computing device 106. The
 display circuit 206 generates and provides a message to the
 teller computing device 106 to display the discrepancy
 information that is determined by the discrepancy circuit
 204. As mentioned above, the display circuit 206 is com-
 municably and operatively coupled to the discrepancy cir-
 cuit 204 to receive identified discrepancy information. As
 mentioned above, the display circuit 206 creates, generates,
 establishes, updates, and maintains a list of potential dis-
 crepancies and any information associated therewith. The
 display includes a list of such information, wherein a teller
 (or any other user) using the teller computing device 106
 may observe or search the list for information associated
 with the potential discrepancies.

A prompt for further action to be displayed on the teller
 computing device is transmitted at 316. In some arrange-
 ments, the display circuit 206 in combination with the
 recommendation circuit 208 transmits the prompt for further
 action to be displayed on the teller computing device 106.
 The recommendation circuit 208 may determine any pos-
 sible further action that a teller may be able to take in regard
 to an identified discrepancy and relay that information to the
 display circuit 206 to transmit to the teller computing device
 106. The display circuit 206 generates and provides a
 message to the teller computing device 106 to display the
 recommended further action. As mentioned above, recom-
 mended actions include, but are not limited to, a manual
 recount of the contents of a drawer device 108 and checking
 other drawer devices for missing or misplaced currency.

Referring now to FIG. 4, an example user interface 400
 showing discrepancy information details for a plurality of
 drawer devices is shown, according to an example embodi-
 ment. The user interface 400 is an example user interface
 that can be presented to a teller via the currency management
 system 146. The user interface 400 allows a teller to view
 potential discrepancies that have occurred during a set time
 period (e.g., the shift of the teller, an hour, a few hours, etc.).

The user interface 400 displays a drawer device number
 402. The drawer device number 402 indicates the drawer
 device that may have a potential discrepancy identified by
 the currency management system 146. As noted above, in
 some embodiments, the teller viewing the user interface 400

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may only see the discrepancy information relating to the
 drawer device for which they are responsible. In other
 embodiments, the user interface 400 displays multiple
 drawer devices 402.

The user interface 400 further displays an expected cur-
 rency amount 404 for each drawer device 402. The expected
 currency amount 404 is determined by the teller transaction
 circuit 202 using information from the transaction database
 148. The expected currency amount 404 is the amount that
 the teller enters into the teller computing device 106 during
 a transaction with the customer. This information may be
 stored in the transaction database 148 for later use in the
 currency management system 146.

The user interface 400 further displays an actual currency
 amount 406 for each drawer device 402. The actual currency
 amount 406 is determined by the teller transaction circuit
 202 using information from the drawer device 108, and in
 particular from the currency counting circuit 128. The
 currency sensor 130 counts the money within a particular
 drawer device 108 (e.g., drawer device 1, drawer device 2,
 etc.) and the currency counting circuit 128 receives that
 information and communicates it to the teller transaction
 circuit 202 within the currency management system 146.
 The actual currency amount 406 is the actual paper money
 and coins the teller receives and places into a drawer device
 during a transaction with the customer. In some embodi-
 ments, the actual currency amount 406 includes a total
 netted amount of currency at the end of a set time period.

For a given drawer device 402, the user interface 400
 further displays a suspected transaction number 408. The
 suspected transaction number 408 indicates the transaction
 during which a potential discrepancy may have occurred.
 The suspected transaction number 408 is determined by the
 discrepancy circuit 204 using information received from the
 teller transaction circuit 202. Additionally, for a given
 drawer device 402, the user interface 400 further provides a
 suspected time 410 indicating the first time at which a
 discrepancy may have occurred. The suspected time 410
 may be related to the suspected transaction number 408 and
 may be used by the currency management system 146 to
 determine during which transaction number 408 a potential
 discrepancy may have occurred.

Referring now to FIG. 5 with reference to FIG. 4, a further
 pop-up window of details on recommended action is shown,
 according to an example embodiment. The pop-up window
 shown in FIG. 5 is displayed if a teller (or other user) selects
 a particular drawer device from the listed items in FIG. 4.
 For example, if a user selects the row for "drawer 1" in FIG.
 4, the user interface 400 opens a further pop-up window
 (e.g., recommended action user interface 500) with further
 recommended action options. If the user selects "drawer 1,"
 the currency management system 146 generates a recom-
 mended action user interface 500 in the form of a pop-up
 window. The recommended action user interface 500
 includes a drawer device number 502 and a list of recom-
 mended actions 504.

As shown, the drawer device number 502 displays the
 name or number of the drawer device 108 selected from the
 user interface 400. Additionally, the list of recommended
 actions 504 displays the options for a user (e.g., teller) to
 take after a potential discrepancy has been identified by the
 currency management system 146. For example, the list of
 recommended actions 504 for Drawer 1 includes "1. Check
 Drawer 2 for Surplus" and "2. Manual Recount" options.
 Thus, in this example, the teller may either decide to go to
 drawer 2 to check if the currency has been misplaced or may
 do a manual recount of the currency. As mentioned with

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regard to FIG. 4, the teller may be aware of the amount of discrepancy such that a recount of the drawer should match that discrepancy (e.g., for drawer 1, the amount of discrepancy was found by the currency management system 146 to be \$110.)

The embodiments described herein have been described with reference to drawings. The drawings illustrate certain details of specific embodiments that implement the systems, methods and programs described herein. However, describing the embodiments with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

It should be understood that no claim element herein is to be construed under the provisions of 35 U.S.C. § 112 (f), unless the element is expressly recited using the phrase “means for.”

As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

The “circuit” may also include one or more dedicated processors communicatively coupled to one or more dedicated memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc.

An example system for implementing the overall system or portions of the embodiments might include a general

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purpose computing computers in the form of computers, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. Each memory device may include non-transient volatile storage media, non-volatile storage media, non-transitory storage media (e.g., one or more volatile and/or non-volatile memories), etc. In some embodiments, the non-volatile media may take the form of ROM, flash memory (e.g., flash memory such as NAND, 3D NAND, NOR, 3D NOR, etc.), EEPROM, MRAM, magnetic storage, hard discs, optical discs, etc. In other embodiments, the volatile storage media may take the form of RAM, TRAM, ZRAM, etc. Combinations of the above are also included within the scope of machine-readable media. In this regard, machine-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions. Each respective memory device may be operable to maintain or otherwise store information relating to the operations performed by one or more associated circuits, including processor instructions and related data (e.g., database components, object code components, script components, etc.), in accordance with the example embodiments described herein.

It should also be noted that the term “input devices,” as described herein, may include any type of input device including, but not limited to, a keyboard, a keypad, a mouse, joystick or other input devices performing a similar function. Comparatively, the term “output device,” as described herein, may include any type of output device including, but not limited to, a computer monitor, printer, facsimile machine, or other output devices performing a similar function.

Any foregoing references to currency or funds are intended to include fiat currencies, non-fiat currencies (e.g., precious metals), and math-based currencies (often referred to as cryptocurrencies). Examples of math-based currencies include Bitcoin, Litecoin, Dogecoin, and the like.

It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied, and the nature or number of discrete processes may be altered or varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative embodiments. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Such variations will depend on the machine-readable media and hardware systems chosen and on designer choice. It is understood that all such variations are within the scope of the disclosure. Likewise, software and web implementations of the present disclosure could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various database searching steps, correlation steps, comparison steps and decision steps.

The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired

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from this disclosure. The embodiments were chosen and described in order to explain the principals of the disclosure and its practical application to enable one skilled in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the embodiments without departing from the scope of the present disclosure as expressed in the appended claims.

What is claimed is:

1. A system comprising:
 - a currency counting device; and
 - a memory and one or more processors communicably coupled to the currency counting device, the memory configured to store instructions that are executable by the one or more processors to cause the one or more processors to:
 - cause a first amount of currency to be counted by the currency counting device, wherein the first amount of currency is an actual amount of currency received by a teller money drawer;
 - determine that the first amount of currency is a discrepancy amount different than an expected amount of currency; and
 - responsive to determining that the discrepancy amount is below a threshold:
 - determine that a teller error is a potential source of the discrepancy; and
 - cause a display device of a teller computing device to display an indication within a pop-up window that the teller error is the potential source of the discrepancy.
2. The system of claim 1, wherein the instructions further cause the one or more processors to cause the display device to display an indication of the discrepancy indicating the discrepancy amount.
3. The system of claim 2, wherein the pop-up window is a second pop-up window, and the instructions further cause the one or more processors to cause the display device to display the indication of the discrepancy indicating the discrepancy amount within a first pop-up window.
4. The system of claim 3, wherein the second pop-up window is nested within the first pop-up window or the first pop-up window is nested in the second pop-up window.
5. The system of claim 1, wherein the instructions further cause the one or more processors to:
 - determine a transaction number identifying a transaction during which the discrepancy occurred,
 - wherein the pop-up window further includes the transaction number.
6. The system of claim 5, wherein the instructions further cause the one or more processors to:
 - determine a transaction time associated with the transaction during which the discrepancy occurred,
 - wherein the pop-up window further includes the transaction time.
7. The system of claim 1, wherein the teller computing device is a mobile device.
8. The system of claim 1, wherein the first amount of currency includes at least one of coins and paper money.
9. A method comprising:
 - counting, by a currency counting circuit, a first amount of currency, wherein the first amount of currency is an actual amount of currency received by a teller money drawer;

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determining, by a discrepancy circuit, that the first amount of currency is a discrepancy amount different than an expected amount of currency; and
responsive to determining that the discrepancy amount is below a threshold:

- determining, by the discrepancy circuit, that a teller error is a potential source of the discrepancy; and
- causing, by a display circuit, a display device of a teller computing device to display an indication within a pop-up window that the teller error is the potential source of the discrepancy.

10. The method of claim 9, wherein the method further comprises causing the display device to display an indication of the discrepancy indicating the discrepancy amount.

11. The method of claim 10, wherein the pop-up window is a second pop-up window, and the method further comprising displaying the indication of the discrepancy indicating the discrepancy amount within a first pop-up window.

12. The method of claim 11, wherein the second pop-up window is nested within the first pop-up window or the first pop-up window is nested in the second pop-up window.

13. The method of claim 9, further comprising:

- determining, by the discrepancy circuit, a transaction number identifying a transaction during which the discrepancy occurred,
- wherein the pop-up window further includes the transaction number.

14. The method of claim 13, further comprising:

- determining, by the discrepancy circuit, a transaction time associated with the transaction during which the discrepancy occurred,
- wherein the pop-up window further includes the transaction time.

15. The method of claim 9, wherein the teller computing device is a mobile device.

16. An apparatus comprising:

- a currency counting circuit configured to count a first amount of currency, wherein the first amount of currency is an actual amount of currency received by a teller money drawer;

- a discrepancy circuit configured to:
 - determine that the first amount of currency is a discrepancy amount different than an expected amount of currency; and

- responsive to determining that the discrepancy amount is below a threshold, determine that a teller error is a potential source of the discrepancy; and

- a display circuit configured to, responsive to the discrepancy circuit determining that the discrepancy amount is below the threshold, cause a display device of a teller computing device to display an indication within a pop-up window that the teller error is the potential source of the discrepancy based on the discrepancy amount being below the threshold.

17. The apparatus of claim 16, wherein the display circuit is further configured to cause the display device to display an indication of the discrepancy indicating the discrepancy amount.

18. The apparatus of claim 17, wherein the pop-up window is a second pop-up window, and the discrepancy circuit is further configured to cause the display device to display the indication of the discrepancy indicating the discrepancy amount within a first pop-up window.

19. The apparatus of claim 18, wherein the second pop-up window is nested within the first pop-up window or the first pop-up window is nested in the second pop-up window.

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20. The apparatus of claim **16**, wherein the teller computing device is a mobile device.

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